

Comparative economic development

Chapter 2

Econ 322 – The Global Economy: Trade and Development
University of Tennessee, Knoxville

August 24, 2026

Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

Overview

- ▶ **How to classify and compare levels of development across countries.**
- ▶ **The diversity within the developing world — and how it compares to the developed world.**
- ▶ **Why global inequalities in income, health, and education persist.**
- ▶ **The concept of convergence vs. divergence: Are poor countries catching up?**

In this chapter

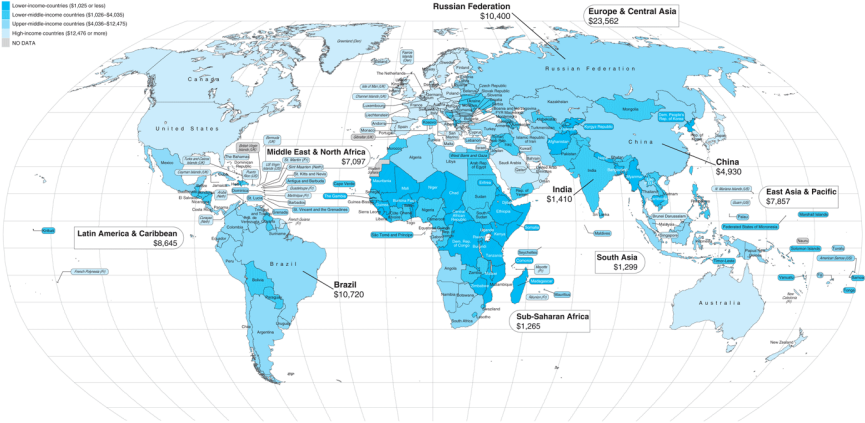
- ▶ **An Introduction**
- ▶ **What is the Developing World? Classifying Levels of National Economic Development**
- ▶ **Comparing Countries by Health and Education, and the Human Development Index**
- ▶ **Key Similarities and Differences Among Developing Countries**
- ▶ **Are Living Standards of Developing and Developed Nations Converging?**
- ▶ **Long-Run Causes of Comparative Development**
- ▶ **Concluding Observations**

What Is the Developing World?

Income

GNI per capita, World Bank Atlas method, 2011

- Lower income-countries (\$1,025 or less)
- Lower middle-income countries (\$1,026-\$4,036)
- Upper middle-income countries (\$4,036-\$12,475)
- High-income countries (\$12,476 or more)
- NO DATA



World map highlighting the developing regions.

Source: *Todaro and Smith (2020), lecture slides*

What Is the Developing World?

- ▶ **Includes most of:**
- ▶ Sub-Saharan Africa
- ▶ Asia
- ▶ North Africa & the Middle East
- ▶ Latin America & the Caribbean
- ▶ Eastern Europe / transition economies
- ▶ Contrasted with the high-income OECD countries (including the U.S., Western Europe, Australia, etc.)

Why Does This Matter?

- ▶ Extreme disparities persist in:
 - ▶ **Income:** GNI per capita (2017): U.S. \$58,270 vs. India \$1,800 vs. DRC \$460
 - ▶ **Health:** life expectancy U.S. 78.7 vs. DRC 59.6 years
 - ▶ **Nutrition:** underweight children India 36% vs. U.S. < 2%
 - ▶ **Literacy:** female literacy U.S. nearly 100%, India 63%

Gaps this large, in income, health, nutrition, and education at once, are what the classification schemes and indices in this chapter (GNI per capita, PPP conversion, the HDI) are designed to measure and compare.

Outline

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- ▶ **What is the Developing World? Classifying Levels of National Economic Development**
- ▶ Comparing Countries by Health and Education, and the Human Development Index
- ▶ Key Similarities and Differences Among Developing Countries
- ▶ Are Living Standards of Developing and Developed Nations Converging?
- ▶ Long-Run Causes of Comparative Development
- ▶ Concluding Observations

Defining the Developing World

- ▶ **The World Bank ranks countries on Gross National Income (GNI) per capita**
- ▶ Low-Income Countries (LICs)
- ▶ Lower-Middle Income Countries (LMCs)
- ▶ Upper-Middle Income Countries (UMCs)
- ▶ High-income OECD countries
- ▶ Other high-income countries
- ▶ (See Table 2.1 and Figure 2.1)

Low-income countries (GNI per capita below \$1,026 in 2018) are concentrated in Sub-Saharan Africa and include fragile or post-conflict states such as South Sudan, Somalia, and Sierra Leone; they are where the development challenge is most urgent.

Table 2.1 Classification of Economies by Country Code, Region, and Income, 2018

Country	Code	Class	Country	Code	Class	Country	Code	Class
East Asia and the Pacific			Latin America and the Caribbean			South Asia		
Cambodia*	KHM	LMC	Bolivia†	BOL	LMC	Afghanistan*†	AFG	LIC
China	CHN	UMC	Brazil	BRA	UMC	Bangladesh*	BGD	LMC
Indonesia	IDN	LMC	Haiti*‡	HTI	LIC	India	IND	LMC
Malaysia	MYS	UMC	Mexico	MEX	UMC	Nepal*†	NPL	LIC
Philippines	PHL	LMC	Nicaragua	NIC	LMC	Pakistan	PAK	LMC
Vietnam	VNM	LMC	Peru	PER	UMC	Sri Lanka	LKA	LMC
Europe and Central Asia			Middle East and North Africa			Sub-Saharan Africa		
Georgia	GEO	LMC	Algeria	DZA	UMC	Congo, Dem. Rep.*	COD	LIC
Kazakhstan†	KAZ	UMC	Egypt, Arab Rep.	EGY	LMC	Ethiopia*†	ETH	LIC
Russian Federation	RUS	UMC	Iran, Islamic Rep.	IRN	UMC	Ghana	GHA	LMC
Tajikistan†	TJK	LIC	Morocco	MAR	LMC	Kenya	KEN	LMC
Turkey	TUR	UMC	Syrian Arab Republic	SYR	LIC	Nigeria	NGA	LMC
Ukraine	UKR	LMC	Yemen, Rep.*	YEM	LIC	South Africa	ZAP	UMC

Selected countries from each region. LIC = low income, LMC = lower-middle income, UMC = upper-middle income. * least-developed; † landlocked developing; ‡ small island developing countries.

Source: Data from World Bank, *World Development Indicators*, 2018

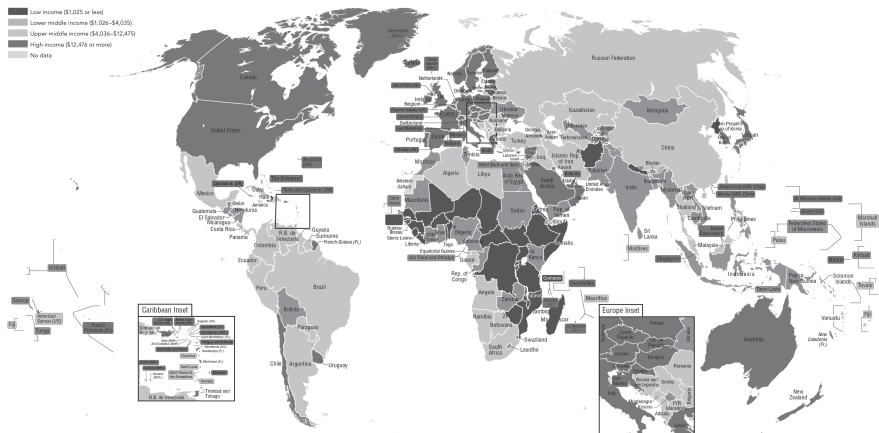
Table 2.1 Classification of Economies by Country Code, Region, and Income, 2018 (Continued)

Country	Code	Country	Code	Country	Code
High-Income OECD Countries				Other High-Income Economies	
Australia	AUS	Chile	CHL	Argentina	ARG
Canada	CAN	Czech Republic	CZE	Bahrain‡	BHR
France	FRA	Israel	ISR	Hong Kong SAR, China	HKG
Germany	DEU	Poland	POL	Kuwait	KWT
Japan	JPN	Portugal	PRT	Qatar	QAT
Korea, Rep.	KOR	Spain	ESP	Saudi Arabia	SAU
United Kingdom	GBR	Sweden	SWE	Singapore‡	SGP
United States	US	Switzerland	CHE	United Arab Emirates	ARE

Selected high-income economies. ‡ small island developing countries.

Source: Data from World Bank, *World Development Indicators*, 2018

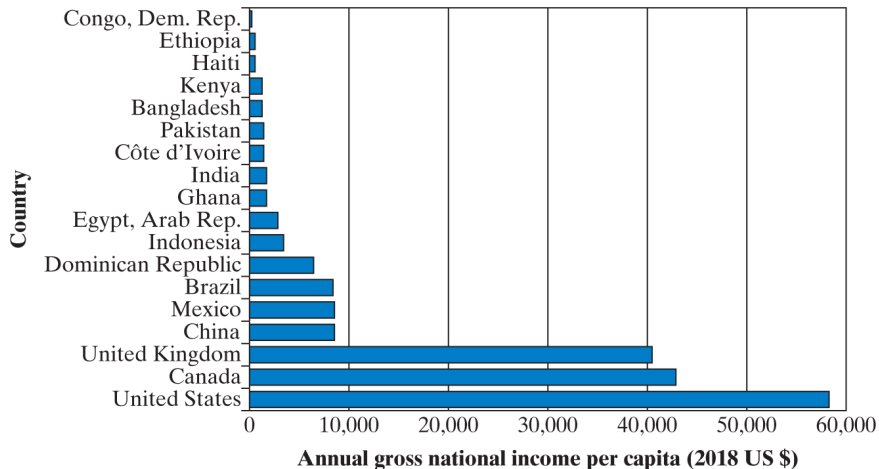
Figure 2.1 Nations of the World, Classified by GNI Per Capita



Countries shaded by World Bank income group.

Source: Todaro and Smith (2020), Figure 2.1; data from World Bank

Figure 2.2 Income Comparisons for Selected Countries, 2017



Source: World Development Indicators

Table 2.2 Per Capita GNI: Exchange Rate vs. PPP Conversion, 2017

Country			Country			Country		
ER	PPP		ER	PPP		ER	PPP	
Bangladesh	1470	4040	Dominican Republic	6630	15290	Pakistan	1580	5830
Bolivia	3130	7340	Egypt, Arab Rep.	3010	11360	Peru	5960	12880
Botswana	6730	16420	Ghana	1880	4280	Philippines	3660	10050
Brazil	8600	15200	Guatemala	4060	8000	Senegal	1240	3360
Cambodia	1230	3750	Haiti	760	1830	Thailand	5950	17040
Canada	42870	46070	India	1800	6980	Uganda	600	1820
Chile	13610	23570	Indonesia	3540	11900	United Kingdom	40530	42560
China	8690	16760	Kenya	1460	3250	United States	58270	60200
Colombia	5890	14090	Korea, Rep.	28380	38340	Vietnam	2160	6450
Congo, Dem. Rep.	460	870	Mexico	8610	17840	Low income	775	2127
Costa Rica	11120	16200	Niger	360	990	Middle income	4942	11993
Côte d'Ivoire	1580	3820	Nigeria	2100	5700	High income	40142	47575

GNI per capita in US \$: ER = official exchange rate conversion, PPP = purchasing power parity conversion.

Source: World Bank World Development Indicators

Basic Indicators of Development: Real Income, Health, and Education

- ▶ Gross National Income (GNI)
- ▶ Gross Domestic Product (GDP)
- ▶ PPP method instead of exchange rates as conversion factors (see Table 2.2)

GNI: total income of resident producers plus net income from abroad.

GDP: total value of goods and services produced within a country; excludes income from abroad.

PPP GNI: GNI converted into international dollars, adjusting for price-level differences across countries; better for comparing living standards.

Table 2.3 Commonality and Diversity: Some Basic Indicators of Health and Education (1/3)

Country	Life exp. at birth (years) 2017	Under-5 mortality (per 1,000) 2018	Under- nourished (% pop.) 2016	Secondary enrolment ratio (%) 2012–17	Some sec. education (% age 25+) 2006–17
Bangladesh	72	30.2	15.2	69	45.5
Bolivia	71	26.8	17.3	86	58.2
Botswana	69	36.5	26.4	–	89.2
Brazil	75	14.4	2.5	100	60
Cambodia	69	28	17.2	–	21.3
Canada	82	5	2.5	113	100
Chile	80	7.2	2.9	100	80.6
China	76	8.6	8.7	95	77.4
Colombia	77	14.2	5.6	98	50.2
Congo, Dem. Rep.	60	88.1	–	46	50.7
Costa Rica	80	8.8	4.7	126	52.9
Côte d'Ivoire	57	80.9	19.6	46	26.1

Source for health indicators: WDI. Source for education indicators: UNDP.

Table 2.3 Commonality and Diversity: Some Basic Indicators of Health and Education (2/3)

Country	Life exp. at birth (years) 2017	Under-5 mortality (per 1,000) 2018	Under- nourished (% pop.) 2016	Secondary enrolment ratio (%) 2012–17	Some sec. education (% age 25+) 2006–17
Dominican Republic	74	28.8	9.9	77	56.6
Egypt, Arab Rep.	72	21.2	4.4	86	64.5
Ghana	63	47.9	5.9	60	62.1
Guatemala	74	26.2	15	64	37.8
Haiti	63	64.8	49	–	33.2
India	69	36.6	14.9	75	51.6
Indonesia	71	25	8.6	86	48.8
Kenya	66	41.1	27.4	–	34.6
Korea, Rep.	83	3.2	2.5	100	95.6
Mexico	75	12.7	3.7	97	59.3
Niger	62	83.7	14.1	24	6.6
Nigeria	54	119.9	11.5	56	–

Source for health indicators: WDI. Source for education indicators: UNDP.

Table 2.3 Commonality and Diversity: Some Basic Indicators of Health and Education (3/3)

Country	Life exp. at birth (years) 2017	Under-5 mortality (per 1,000) 2018	Under- nourished (% pop.) 2016	Secondary enrolment ratio (%) 2012–17	Some sec. education (% age 25+) 2006–17
Pakistan	67	69.3	20.6	46	37.3
Peru	76	14.3	9.7	98	62.2
Philippines	71	28.4	13.5	88	73.2
Senegal	67	43.6	12	48	17.1
Thailand	77	9.1	7.8	121	44.8
Uganda	63	46.4	39.7	–	31.7
United Kingdom	81	4.3	2.5	125	82.9
United States	79	6.5	2.5	97	95.3
Vietnam	75	20.7	9.4	–	69.4
Low income	63	68.1	68.1		
Lower middle income	68	49.1	49.1		
Upper middle income	76	12.6	12.6		
High income	81	5	5		

Source for health indicators: WDI. Source for education indicators: UNDP.

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Holistic Measures of Living Levels and Capabilities

- ▶ Income is one indicator, but needs to be supplemented with others
- ▶ **Health**, e.g., Life Expectancy
- ▶ **Education**
- ▶ Other indicators are considered in various indices
- ▶ The **New Human Development Index (NHDI)**, or simply “HDI”
- ▶ Introduced by UNDP in November 2010
- ▶ NHDI, as an attempt to create and use a holistic measure of living levels, takes into account income, health, and education
- ▶ NHDI can be calculated for groups and regions in a country
 - ▶ HDI varies among groups within countries
 - ▶ HDI varies across regions in a country
 - ▶ HDI varies between rural and urban areas

Why the New HDI Uses a Geometric Mean

- ▶ How does the New HDI compare with the better-known (but no longer active) Traditional HDI? And other linear combinations of national indexes?
- ▶ Probably most consequential: The HDI is now computed with a geometric mean, instead of an arithmetic mean
- ▶ A geometric mean is also used to build up the overall education index from its two components
- ▶ Traditional HDI added the three components, divided by 3 (arithmetic mean)
- ▶ The New HDI takes the cube root of the product of the 3 component indexes
- ▶ The traditional HDI linear calculation assumed one component traded off against another as perfect substitutes, a strong assumption
- ▶ The reformulation now allows for imperfect substitutability - widely considered a more plausible way to frame the tradeoffs

Calculating with a geometric mean in the New HDI

- ▶ The New HDI takes the geometric mean, i.e. the cube root of the product of the three component indexes I_{life} , $I_{\text{education}}$, and I_{income} :

$$\text{NHDI} = I_{\text{life}}^{1/3} I_{\text{education}}^{1/3} I_{\text{income}}^{1/3} = (I_{\text{life}} I_{\text{education}} I_{\text{income}})^{1/3}$$

- ▶ This reformulation allows for imperfect substitutability
- ▶ Addresses “how well-rounded” a country’s performance is across the three dimensions.”
- ▶ Other differences (choices) made in the components of the sub-indices
- ▶ We can see these elements in the textbook’s Example of Ghana:

Calculating the New HDI: Example (Ghana)

Indicator	Value
Life expectancy at birth (years)	64.6
Mean years of schooling	7.0
Expected years of schooling	11.4
GNI per capita (PPP \$)	1,684

- ▶ Life expectancy index = $\frac{64.6 - 20}{83.6 - 20} = 0.701$
- ▶ Mean years sub-index = $\frac{7.0 - 0}{13.3 - 0} = 0.527$
- ▶ Expected years sub-index = $\frac{11.4 - 0}{18.0 - 0} = 0.634$
- ▶ Education index = $\frac{\sqrt{0.527 \times 0.634} - 0}{0.971 - 0} = 0.596$
- ▶ Income index = $\frac{\ln 1,684 - \ln 100}{\ln 87,478 - \ln 100} = 0.417$

$$\text{NHDI} = 0.701^{1/3} \times 0.596^{1/3} \times 0.417^{1/3} = 0.558$$

Note: Example from the 12th edition; numbers are rounded. Source:
http://hdr.undp.org/sites/default/files/hdr_2013_en_technotes.pdf

Other key differences between the New HDI and Traditional HDI

- ▶ **Traditional HDI is still widely taught; students may encounter it in other classes**
- ▶ **A brief summary of other key differences between the New HDI and the Traditional HDI (in addition to using a geometric mean) follows:**
- ▶ **Gross national income** per capita replaces **gross domestic product** per capita
- ▶ Revised education components: the New HDI uses the average actual educational attainment of the whole population, and the expected attainment of today's children (not enrollment)
- ▶ The maximum values in each dimension have been increased to the **observed maximum** rather than given a **predefined cutoff**
- ▶ Note: Please be sure to review example country comparisons to get a sense of how much HDI rankings can differ from income rankings

Table 2.4 2018 Human Development Index and its Components for Selected Countries (1/2)

Country	HDI rank	Life exp. at birth	Mean yrs of schooling	Expected yrs of schooling	GNI per capita	HDI value	GNI p.c. rank minus HDI rank
Canada	12	82.5	13.3	16.4	43,433	0.926	10
United States	13	79.5	13.4	16.5	54,941	0.924	-2
United Kingdom	14	81.7	12.9	17.4	39,116	0.922	13
South Korea	22	82.4	12.1	16.5	35,945	0.903	8
United Arab Emirates	34	77.4	10.8	13.6	67,805	0.863	-27
Chile	44	79.7	10.3	16.4	21,910	0.843	13
Russian Federation	49	71.2	12.0	15.5	24,233	0.816	3
Costa Rica	63	80.0	8.8	15.4	14,636	0.794	15
Turkey	64	76.0	8.0	15.2	24,804	0.791	-14
Cuba	73	79.9	11.8	14.0	7,524	0.777	43
Mexico	74	77.3	8.6	14.1	16,944	0.774	-6
Sri Lanka	76	75.5	10.9	13.9	11,326	0.770	19
Brazil	79	75.7	7.8	15.4	13,755	0.759	2
China	86	76.4	7.8	13.8	15,270	0.752	-9
Botswana	101	67.6	9.3	12.6	15,534	0.717	-26
Gabon	110	66.5	8.2	12.8	16,431	0.702	-40

Source: United Nations Development Program

Table 2.4 2018 Human Development Index and its Components for Selected Countries (2/2)

Country	HDI rank	Life exp. at birth	Mean yrs of schooling	Expected yrs of schooling	GNI per capita	HDI value	GNI p.c. rank minus HDI rank
South Africa	113	63.4	10.1	13.3	11,923	0.699	-23
Egypt	115	71.7	7.2	13.1	10,355	0.696	-15
Guatemala	127	73.7	6.5	10.8	7,278	0.650	-8
India	130	68.8	6.4	12.3	6,353	0.640	-5
Bangladesh	136	72.8	5.8	11.4	3,677	0.608	9
Ghana	140	63.0	7.1	11.6	4,096	0.592	3
Equatorial Guinea	141	57.9	5.5	9.3	19,513	0.591	-80
Kenya	142	67.3	6.5	12.1	2,961	0.590	16
Pakistan	150	66.6	5.2	8.6	5,311	0.562	-14
Papua New Guinea	153	65.7	4.6	10.0	3,403	0.544	-3
Madagascar	161	66.3	6.1	10.6	1,358	0.519	20
Côte d'Ivoire	170	54.1	5.2	9.0	3,481	0.492	-22
Burkina Faso	183	60.8	1.5	8.5	1,650	0.423	-7
Chad	186	53.2	2.3	8.0	1,750	0.404	-15
Niger	189	60.4	2.0	5.4	906	0.354	-2

Source: United Nations Development Program

Table 2.5 HDI for Countries with Similar Income Levels (1/3)

Country	HDI value (2015)	Life exp. at birth	Expected yrs of schooling	Average yrs of schooling	GNI per capita	GNI p.c. rank minus HDI rank
Country GDP per capita near PPP \$1000						
158-Madagascar	0.512	65.5	10.3	6.1	1,320	25
166-Togo	0.487	60.2	12.0	4.7	1,262	18
170-Malawi	0.476	63.9	10.8	4.4	1,073	16
183-Guinea	0.414	59.2	8.8	2.6	1,058	4
187-Niger	0.353	61.9	5.4	1.7	889	1
Country GDP per capita near PPP \$3500						
139-Bangladesh	0.579	72.0	10.2	5.2	3,341	8
139-Zambia	0.579	60.8	12.5	6.9	3,464	7
157-Mauritania	0.513	63.2	8.5	4.3	3,527	-12
160-Lesotho	0.497	50.1	10.7	6.1	3,319	-12
165-Sudan	0.490	63.7	7.2	3.5	3,846	-22

Data source: Human Development Report 2016, Table 1, pages 198–201 (New York: United Nations Development Program), 2015 data.

Table 2.5 HDI for Countries with Similar Income Levels (2/3)

Country	HDI value (2015)	Life exp. at birth	Expected yrs of schooling	Average yrs of schooling	GNI per capita	GNI p.c. rank minus HDI rank
Country GDP per capita near PPP \$7500						
68-Cuba	0.775	79.6	13.9	11.8	7,455	48
84-Ukraine	0.743	71.1	15.3	11.3	8,189	28
103-Belize	0.706	70.1	12.8	10.5	7,375	14
117-El Salvador	0.680	73.3	13.2	6.5	7,732	-3
148-Eswatini (Swaziland)	0.541	48.9	11.4	6.8	7,522	-33
Country GDP per capita near PPP \$10,000						
73-Sri Lanka	0.766	75.0	14.0	10.9	10,789	21
86-Jordan	0.741	74.2	13.1	10.1	10,111	15
111-Egypt	0.691	71.3	13.1	7.1	10,064	-7
113-Indonesia	0.689	69.1	12.9	7.9	10,053	-8
125-Namibia	0.640	65.1	11.7	6.7	9,770	-18

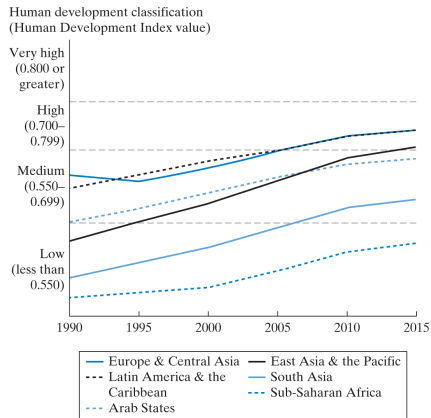
Data source: Human Development Report 2016, Table 1, pages 198–201 (New York: United Nations Development Program), 2015 data.

Table 2.5 HDI for Countries with Similar Income Levels (3/3)

Country	HDI value (2015)	Life exp. at birth	Expected yrs of schooling	Average yrs of schooling	GNI per capita	GNI p.c. rank minus HDI rank
Country GDP per capita near PPP \$20,000						
54-Uruguay	0.795	77.4	15.5	8.6	19,148	8
60-Panama	0.788	77.8	13.0	9.9	19,470	0
109-Gabon	0.697	64.9	12.6	8.1	19,044	−46
135-Equatorial Guinea	0.592	57.9	9.2	5.5	21,517	−79

Data source: Human Development Report 2016, Table 1, pages 198–201 (New York: United Nations Development Program), 2015 data.

Figure 2.3 Improvements in Human Development Since 1990, by Region



Source: Human Development Report Office, UNDP – Human Development Report, 2016, p. 27

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Comparing characteristics among developing countries

- ▶ **Ten points of comparison - both among developing countries, and between developing and developed countries:**
- ▶ Lower levels of living and productivity
- ▶ Lower levels of human capital
- ▶ Higher levels of inequality and absolute poverty
- ▶ Higher population growth rates
- ▶ Greater social fractionalization
- ▶ Larger rural population - rapid migration to cities
- ▶ Lower levels of industrialization and manufactured exports
- ▶ Adverse geography
- ▶ Underdeveloped financial and other markets
- ▶ Colonial Legacies – quality of institutions

Comparing developing countries (1): Income and productivity

- ▶ **1. Levels of income and productivity**
- ▶ **Large productivity and income gaps** exist between developed and developing countries
 - ▶ Also wide variation **within** developing countries (e.g., India vs. DRC)
- ▶ **Main driver:** differences in **output per worker**
 - ▶ Low income → low investment (health, education, infrastructure) → low productivity → stagnation
 - ▶ This cycle is known as a **poverty trap**
- ▶ **Escaping low income is possible**
 - ▶ Success stories: **South Korea, Taiwan, China, India**
- ▶ **Growth patterns vary by region**
 - ▶ Rapid: East Asia; Slow: Sub-Saharan Africa; Mixed: Latin America, others
- ▶ **Country size $\neq$ development outcome**
 - ▶ No clear link between population/land size and income

Table 2.6 The 12 Most- and Least-Populated Countries and Their Per Capita Income, 2017

Most populous	Population (millions)	GNI p.c. (US \$)	Least populous	Population (thousands)	GNI p.c. (US \$)
1. China	1386	8690	1. Tuvalu	11	4970
2. India	1339	1800	2. Nauru	14	10220
3. United States	326	58270	3. Palau	22	12700
4. Indonesia	264	3540	4. Marshall Islands	53	4840
5. Brazil	209	8600	5. St. Kitts and Nevis	55	16240
6. Pakistan	197	1580	6. Dominica	74	6590
7. Nigeria	191	2100	7. Seychelles	96	14170
8. Bangladesh	165	1470	8. Antigua and Barbuda	102	13810
9. Russian Federation	144	9230	9. Micronesia	106	3620
10. Mexico	129	8610	10. Grenada	108	9180
11. Japan	127	38550	11. Tonga	108	4010
12. Ethiopia	105	740	12. St. Vincent and the Grenadines	110	7390

Source: World Bank World Development Indicators

Comparing developing countries (2): Human capital

▶ 2. Human Capital Attainments

▶ **Why it matters:** health, education, and skills are key to growth and development.

▶ Progress and disparities

- ▶ Major gains in past 25 years
- ▶ But large gaps remain—especially in the poorest countries
- ▶ Under-5 mortality still $\sim 15\times$ higher in low-income countries

▶ Education challenges

- ▶ Primary enrollment rising, but learning quality lags
- ▶ Problems: low attendance, weak basic skills, high pupil-teacher ratios
- ▶ Teacher absenteeism is widespread in some regions

▶ Health conditions

- ▶ Upper-middle-income countries nearing high-income levels
- ▶ Sub-Saharan Africa and South Asia still face major health and nutrition challenges

Comparing developing countries (3): Inequality and poverty

▶ 3. Inequality and Absolute Poverty

▶ Inequality across countries

- ▶ High in Latin America, Sub-Saharan Africa, and resource-rich nations
- ▶ Generally lower (but rising) in Asia

▶ Extreme poverty

- ▶ Defined as living on $< \$1.90/\text{day}$
- ▶ Concentrated in Sub-Saharan Africa and South Asia
- ▶ Linked to low human capital, exclusion, and conflict

▶ Progress and remaining gaps

- ▶ Poverty rate fell from 42% (1981) to 10% (2017)
- ▶ But 46% still live on $< \$5.50/\text{day}$

Comparing developing countries (4): Population growth and age structure

- ▶ **4. Population Growth and Age Structure**
- ▶ **Global population trends:** grew from < 1 billion (1800) → 1.65 billion (1900) → 6 billion (2000) → 7.6 billion (2018); growth rate declined globally: ~1.1% in 2018
- ▶ **Regional growth patterns**
 - ▶ Rapid growth concentrated in **low-income countries**, especially **Sub-Saharan Africa** (2.7% annually, 2000–2017)
 - ▶ Lower growth in **South Asia** (1.5%), **Latin America** (1.2%), **East Asia** (0.7%)
- ▶ **Fertility rates:** fell globally from **5 births/woman (1965)** to **2.5 (2015)**; still high in many low-income countries
- ▶ **Dependency and age structure:** low-income countries have a large **youth dependency burden**; high-income countries a rising **elderly burden**
- ▶ **Demographic transition and dividend**
 - ▶ Many middle-income countries now have **lower birth rates** and **large working-age populations**: a “**demographic dividend**” for growth
 - ▶ But aging risk looms: e.g., **China’s workforce is already shrinking**

Comparing developing countries (5): Rural economy and migration (1/2)

- ▶ **5. Rural Economy and Rural-to-Urban Migration**
- ▶ **Rural dominance in developing countries**
 - ▶ Many low- and middle-income countries still have **majority rural populations**
 - ▶ Especially true in **Sub-Saharan Africa, South Asia, and Southeast Asia**
- ▶ **Rural challenges**
 - ▶ Rural areas are generally **poorer**
 - ▶ Face **missing markets, limited information, and social barriers**
 - ▶ Slower access to education, health, and infrastructure

Comparing developing countries (5): Rural economy and migration (2/2)

► Urbanization trends

- Global population became majority **urban** around **2012**
- Massive rural-to-urban migration now underway
- Urbanization brings **both opportunities and new challenges**

► Development implications

- Urban growth often outpaces planning and services
- Managing migration and improving rural conditions are both critical for balanced development



Lagos, Nigeria, one of the fastest-growing megacities in the world.

Wikimedia Commons, kopee15, Public domain

Comparing developing countries (6): Social fractionalization (1/2)

▶ 6. Social Fractionalization

▶ Definition and prevalence

- ▶ **Fractionalisation:** ethnic, linguistic, or religious divisions within a country
- ▶ More common in **low-income** countries; can be linked to **conflict**, **governance issues**, and **slow development**

▶ Development impacts

- ▶ High diversity often correlates with **lower growth**, **weaker education outcomes**, **political instability**, and **poor infrastructure**
- ▶ Risk increases when **inequality overlaps with identity lines**

Comparing developing countries (6): Social fractionalization (2/2)

► **Conflict and exclusion**

- Over half of developing countries have experienced **interethnic conflict**
- Discrimination and exclusion harm groups like **indigenous peoples** in Latin America, **ethnic minorities** in Africa and Asia, and **descendants of African slaves** in the Americas

► **Diversity $\neq$ destiny**

- Diversity can be a **source of innovation** (e.g., U.S., Malaysia, Mauritius)
- Outcomes depend on **institutions, inclusion, and social cohesion**

Comparing developing countries (7): Industrialization and exports (1/2)

- ▶ **7. Level of Industrialization and Manufactured Exports**
- ▶ **Industrialisation and development**
 - ▶ Tied to **high productivity** and **national economic strength**; many developing countries prioritize **manufacturing-led growth**, with **notable success** in Asia
- ▶ **Sectoral employment and output**
 - ▶ **Developed countries:** agriculture < 2% of GDP and employment, with high productivity
 - ▶ **Low-income countries:** agriculture = majority employment, but low productivity (Madagascar: 82% in agriculture, 25% of output; Indonesia: 41% and 14%)

Comparing developing countries (7): Industrialization and exports (2/2)

- ▶ **Gender and sectoral shifts**
 - ▶ In **Africa and Asia**, many **women** work in agriculture; **developed economies** shifted from **industry to services**, while industry still employs many in **developing countries**
- ▶ **Export composition**
 - ▶ **Middle-income countries**: rising share of **manufactured exports**
 - ▶ **Low-income countries**: still rely on **agriculture and minerals**; manufacturing often **low-skill and low-tech**

Comparing developing countries (8): Geography and resources (1/2)

▶ 8. Geography and Natural Resource Endowments

▶ Geographic disadvantages

- ▶ **Landlocked countries** often have **lower income** (common in Africa)
- ▶ **Tropical regions** face endemic diseases (e.g., malaria), agricultural pests, water shortages, high heat stress

▶ Climate change risk

- ▶ **Sub-Saharan Africa** and **South Asia** face greatest projected negative impacts
- ▶ Many poor countries lack resources to adapt

Comparing developing countries (8): Geography and resources (2/2)

▶ **Natural resource endowments**

- ▶ Resource-rich countries (e.g., Persian Gulf states) can benefit
- ▶ But “**resource curse**”: mineral wealth often leads to conflict, inequality, and weak governance (example: Democratic Republic of Congo)

▶ **Geography $\neq$ destiny**

- ▶ Some tropical regions are highly successful: **Singapore, Southern India**

Comparing developing countries (9): Market development (1/2)

- ▶ **9. Extent of Financial and Other Market Development**
- ▶ **Market imperfections in developing countries**
 - ▶ **Imperfect markets** and **incomplete information** are widespread
 - ▶ Lead to **misallocation** of resources and reduced productivity
- ▶ **Institutional weaknesses** — many developing countries lack:
 - ▶ Effective **legal systems** (contracts, property rights)
 - ▶ Stable **currencies** and macroeconomic environments
 - ▶ Reliable **infrastructure** for trade and communication
 - ▶ Inclusive and regulated **financial systems** (banks, credit, insurance)
 - ▶ Access to **market information** and creditworthiness data
 - ▶ **Social norms** that support long-term trust and contracts

Comparing developing countries (9): Market development (2/2)

► **Broader issues**

- **Thin markets, externalities, and poorly regulated common resources** (e.g., land, water)
- Can lead to **underdevelopment traps**

Without working legal systems, credit markets, and information, resources are misallocated and productivity stays low; the market imperfections themselves can therefore trap an economy in underdevelopment.

Comparing developing countries (10): Institutions and dependence (1/2)

- ▶ **10. Quality of Institutions and External Dependence**
- ▶ **Lasting impacts of colonialism**
 - ▶ Most developing countries were former colonies; institutions focused on **extraction**, not **wealth creation**
 - ▶ Legacy: weak property rights, elite control, exclusion, and inequality
- ▶ **Institutional and social consequences**
 - ▶ **Forced labor, slavery, and ethnic divisions** had long-term effects
 - ▶ **High inequality** hindered democracy, education, investment; elites replicated colonial patterns

Comparing developing countries (10): Institutions and dependence (2/2)

▶ **External dependence**

- ▶ **Weak bargaining power** in trade; WTO rules, IP protection, global crises favored developed nations
- ▶ Reliance on developed countries for finance, technology, and environmental stability

▶ **Environmental and cultural dependence**

- ▶ Climate change hits developing nations hardest, yet emissions come mainly from rich countries
- ▶ Western cultural and media influence also shapes development pathways

Outline

- ▶ An Introduction
- ▶ What is the Developing World? Classifying Levels of National Economic Development
- ▶ Comparing Countries by Health and Education, and the Human Development Index
- ▶ Key Similarities and Differences Among Developing Countries
- ▶ **Are Living Standards of Developing and Developed Nations Converging?**
- ▶ Long-Run Causes of Comparative Development
- ▶ Concluding Observations

The Great Divergence (1/2)

► The Great Divergence

- In 1750, richest countries were only $\sim 3\times$ richer than the poorest; today the ratio is closer to **100:1**
- Divergence occurred over **two centuries of industrial-led growth** in Europe and North America
- Many others—especially colonies—**stagnated or declined**

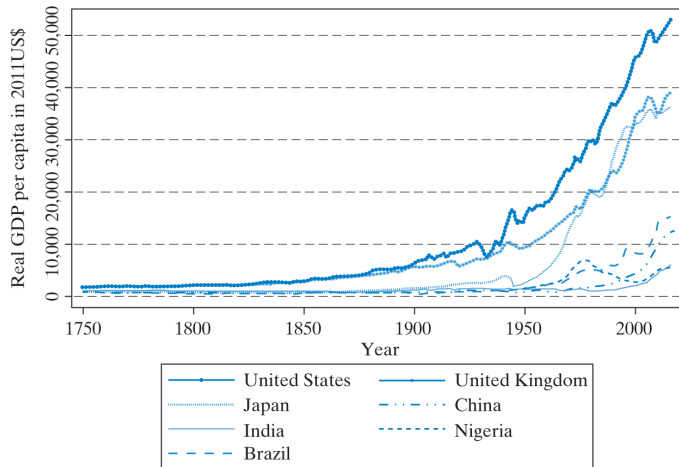
► Causes of divergence

- **Industrial Revolution:** steam, mechanization, productivity boom
- **Colonial extraction:** resource plunder, industrial suppression in colonies
- **Slave-based systems:** supported capital accumulation in Europe and the U.S.

The Great Divergence (2/2)

- ▶ **Convergence beginnings**
 - ▶ **Japan:** first non-Western country to catch up
 - ▶ **China and India:** began sustained growth in the 1990s
 - ▶ Many low-income countries **still lag behind**
- ▶ **Key question:** are global living standards now **converging**? Or is divergence still shaping development outcomes?

Figure 2.5 The growth of real output per person since 1750



Source: Data from Maddison Project Database

Two Major Reasons to Expect Convergence

▶ 1. Technology transfer

- ▶ Developing countries can **adopt existing technologies** rather than reinvent them
- ▶ Can **leapfrog** earlier stages of development
- ▶ Known as the “**advantage of backwardness**” (Gerschenkron)
- ▶ Example: time to double output per worker — UK: 60 years; South Korea: 12 years; China: 8 years

▶ 2. Diminishing returns to capital

- ▶ Developed countries: **high capital stock** → **lower marginal returns**
- ▶ Developing countries: **higher returns** on new investment
- ▶ Should attract both **domestic and foreign investment**

Country-Level Perspectives on Income Convergence

▶ Relative income convergence

- ▶ 1970–1994: only 63 of 152 countries grew faster than the U.S. → **divergence**
- ▶ 1994–2017: 116 countries grew faster than the U.S. → **re-convergence** begins

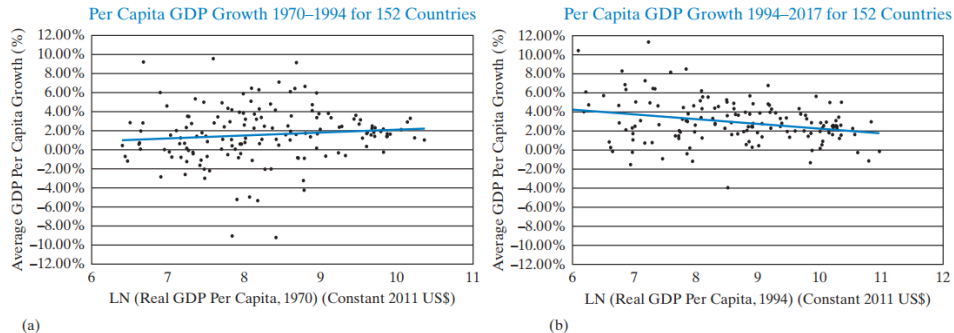
▶ Conditional convergence

- ▶ Based on Solow growth theory
- ▶ Countries **converge in income levels** only **if** savings rates, population growth, and productivity are similar
- ▶ A **weaker** but more realistic form of convergence

▶ Selection bias warning

- ▶ Early studies using only OECD countries showed convergence
- ▶ But this was **biased**—developing countries often **diverged** during 1965–2005
- ▶ Always ask: **which countries are included in the sample?**

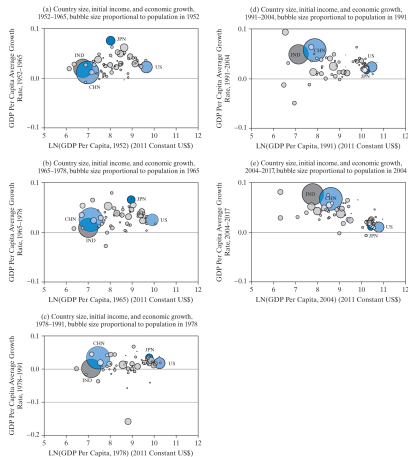
Figure 2.6 Relative Country Convergence 1970–1994 and 1994–2017



Growth of GDP per capita relative to the U.S. in the two sub-periods.

Source: *Todaro and Smith (2020), Figure 2.6; data from Penn World Table*

Figure 2.7 Relative Country Convergence: World, Developing Countries, and OECD



Convergence patterns differ by sample: world, developing countries, OECD.

Source: *Todaro and Smith (2020), Figure 2.7; data from Penn World Table*

Broader Perspectives on Convergence

- ▶ **Population-weighted convergence**
 - ▶ Larger countries (China, India) drive recent convergence trends; especially strong between **2004–2017**
 - ▶ More realistic than treating each country equally
- ▶ **Absolute vs. relative convergence**
 - ▶ **Relative:** % income growth higher in poorer countries
 - ▶ **Absolute:** actual income gains still **larger** in rich countries (China's GDP grew fast, but U.S. absolute income gain was still bigger)
- ▶ **Global (world-as-one-country) view**
 - ▶ Uses **individuals** rather than countries as units
 - ▶ Inequality among people peaked (1988–1993), now **declining**; still rising **within-country inequality** in places like China, India, U.S.
- ▶ **Future of convergence: not guaranteed**
 - ▶ Risks: tech divides, climate change, policy failures, conflict
 - ▶ But momentum is growing: we may be entering a **Great Re-Convergence**

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- ▶ **Long-Run Causes of Comparative Development**
- ▶ Concluding Observations

Long-Run Causes: 1. Geography

► 1. Geography – Initial Conditions

- Geography includes climate, location, disease burden (e.g., malaria), and access to the sea.
- Once considered a major factor, but now seen as secondary to institutions.
- Still relevant in shaping early institutions and production patterns (e.g., Arrow 1).
- Examples: landlocked countries may face higher trade costs; malaria remains a barrier to human capital development.

Is geography destiny? Modern research says geography matters mostly *indirectly*: it influenced the kinds of colonies that were established, which in turn shaped institutions.

Long-Run Causes: 4-5. Precolonial Structures and Inequality

▶ 4. Precolonial Structures and Comparative Advantage

- ▶ Engerman and Sokoloff: local conditions shaped colonial institutions.
- ▶ Sugarcane + slavery → extreme inequality.
- ▶ Grain economies + labor scarcity → egalitarian development.
- ▶ The resulting inequality affected education, land access, and voting (Arrows 6–7).

▶ 5. Inequality as a Development Constraint

- ▶ High inequality limits: human capital formation; public goods provision; democratic development (Arrows 11–13).
- ▶ Easterly: land endowments predict inequality; inequality reduces income.

Long-Run Causes: 6. Human Capital

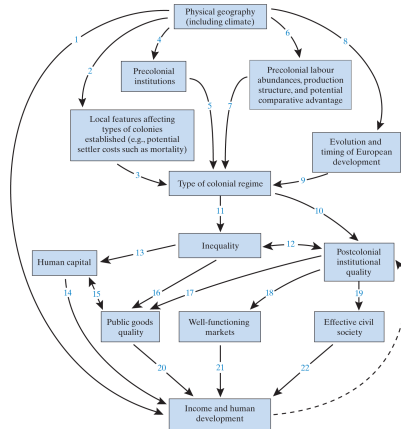
▶ 6. Human Capital and Institutional Feedback

- ▶ Education enhances productivity, civic awareness, and government quality.
- ▶ Institutions and human capital reinforce each other (Arrow 14).
- ▶ Example: Banerjee & Iyer on colonial India — landlord systems led to lower long-run investment.

▶ 7. Role of Civil Society and Governance

- ▶ Democratic governance promotes accountability and broad-based public goods (Arrow 17).
- ▶ Civil society supports inclusive policy-making (Arrow 19).
- ▶ Dani Rodrik: democracy as a “meta-institution” for building better institutions.

Figure 2.9 Schematic Representation of Leading Theories of Comparative Development



The numbered arrows referenced on the preceding frames.

Source: Todaro and Smith (2020), Figure 2.9

Chapter summary

- ▶ The World Bank groups countries by GNI per capita (low, lower-middle, upper-middle, high income); PPP conversion narrows, but does not remove, the income gaps.
- ▶ The New HDI combines health, education, and income with a geometric mean, so countries with similar incomes can have very different human development.
- ▶ Developing countries share ten broad characteristics (lower productivity, human capital, higher inequality and population growth, rural populations, less industrialization, adverse geography, weak markets, colonial legacies) but differ widely.
- ▶ After two centuries of divergence, relative convergence has begun since the mid-1990s, driven by large countries such as China and India; it is not guaranteed.
- ▶ Long-run causes: geography acts mostly through institutions; colonial institutions and inequality persist and shape human capital, governance, and growth.

Appendixes

Appendix: Holistic Measures of Living Levels and Capabilities - The Traditional HDI

- ▶ Income is one indicator, but supplement with others
- ▶ Health, e.g. Life Expectancy
- ▶ Education, e.g., school enrollment levels
- ▶ Other indicators can be and are considered in various indices
- ▶ Traditional HDI was developed in 1990 as a holistic measure of living levels; takes into account income, health, and education
- ▶ It is still used by some as an alternative to the New HDI
- ▶ What many still think of as the current HDI, so worth knowing
- ▶ HDI can be calculated for groups and regions in a country
- ▶ HDI varies among groups within countries
- ▶ HDI varies across regions in a country
- ▶ HDI varies between rural and urban areas

Holistic Measures of Living Levels and Capabilities: Traditional HDI

- ▶ From the Appendix: Traditional HDI
- ▶ A detailed calculation (example of Bangladesh) is found in Appendix 2.1
- ▶ Unlike the New HDI, the Traditional HDI is calculated using an arithmetic mean:

$$\text{HDI} = \frac{1}{3} (\text{income index}) + \frac{1}{3} (\text{life expectancy index}) + \frac{1}{3} (\text{education index})$$

Table A2.1.1 2009 Traditional Human Development Index for 24 Selected Countries (2007 Data) (1/3)

Country	HDI ranking	Human Development Index (HDI)	GDP per capita (PPP, US \$)	GDP per capita rank minus HDI rank
Low Human Development				
Niger	182	0.340	627	−6
Afghanistan	181	0.352	1,054	−17
Dem. Rep. Congo	176	0.389	298	5
Ethiopia	171	0.414	779	0
Rwanda	167	0.460	866	1
Côte d'Ivoire	163	0.484	1,690	−17
Malawi	160	0.493	761	12

Source: Data from United Nations Development Programme, Human Development Report, 2009, tab. 1.

Table A2.1.1 2009 Traditional Human Development Index for 24 Selected Countries (2007 Data) (2/3)

Country	HDI ranking	Human Development Index (HDI)	GDP per capita (PPP, US \$)	GDP per capita rank minus HDI rank
Medium Human Development				
Bangladesh	146	0.543	1,241	9
Pakistan	141	0.572	2,496	−9
India	134	0.612	2,753	−6
South Africa	129	0.683	9,757	−51
Nicaragua	124	0.699	2,570	6
Gabon	103	0.755	15,167	−49
China	92	0.772	5,383	10
Iran	88	0.782	10,955	−17
Thailand	87	0.783	8,135	−5

Source: Data from United Nations Development Programme, Human Development Report, 2009, tab. 1.

Table A2.1.1 2009 Traditional Human Development Index for 24 Selected Countries (2007 Data) (3/3)

Country	HDI ranking	Human Development Index (HDI)	GDP per capita (PPP, US \$)	GDP per capita rank minus HDI rank
High Human Development				
Saudi Arabia	59	0.843	22,935	-19
Costa Rica	54	0.854	10,842	19
Cuba	51	0.863	6,876	44
Chile	44	0.878	13,880	15
Very High Human Development				
United Kingdom	21	0.947	35,130	-1
United States	13	0.956	45,592	-4
Canada	4	0.966	35,812	14
Norway	1	0.971	53,433	4

Source: Data from United Nations Development Programme, Human Development Report, 2009, tab. 1.

Table A2.1.2 2009 HDI Variations for Similar Incomes (2007 Data) (1/2)

Country	GDP per capita (US \$)	HDI	HDI rank	Life expectancy (years)	Adult literacy (%)	Combined gross enrolment ratio
GDP per capita near PPP \$1,000						
Madagascar	932	0.543	145	59.9	70.7	61.3
Haiti	1,140	0.532	149	61.0	62.1	52.1
Rwanda	866	0.460	167	49.7	64.9	52.2
Mali	1,083	0.371	178	48.1	26.2	46.9
Afghanistan	1,054	0.352	181	43.6	28.0	50.1
GDP per capita near PPP \$1,500						
Kenya	1,542	0.541	147	53.6	73.6	59.6
Ghana	1,334	0.526	152	56.5	65.0	56.5
Côte d'Ivoire	1,690	0.484	163	56.8	48.7	37.5
Senegal	1,666	0.464	166	55.4	41.9	41.2
Chad	1,477	0.392	175	48.6	31.8	36.5

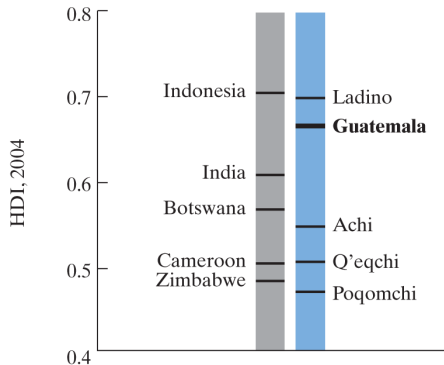
Source: Data from United Nations Development Programme, Human Development Report, 2009, tab. 1.

Table A2.1.2 2009 HDI Variations for Similar Incomes (2007 Data) (2/2)

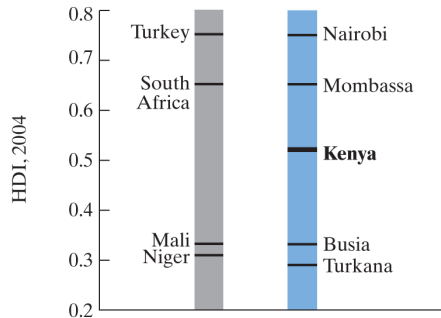
Country	GDP per capita (US \$)	HDI	HDI rank	Life expectancy (years)	Adult literacy (%)	Combined gross enrolment ratio
GDP per capita near PPP \$2,000						
Kyrgyzstan	2,006	0.710	120	67.6	99.3	77.3
Laos	2,165	0.619	133	64.6	72.7	59.6
Cambodia	1,802	0.593	137	60.6	76.3	58.5
Sudan	2,086	0.531	150	57.9	60.9	39.9
Cameroon	2,128	0.523	153	50.9	67.9	52.3
Mauritania	1,927	0.520	154	56.6	55.8	50.6
Nigeria	1,969	0.511	158	47.7	72.0	53.0
GDP per capita near PPP \$4,000						
Tonga	3,748	0.768	99	71.7	99.2	78.0
Sri Lanka	4,243	0.759	102	74.0	90.8	68.7
Honduras	3,796	0.732	112	72.0	83.6	74.8
Bolivia	4,206	0.729	113	65.4	90.7	86.0
Guatemala	4,562	0.704	122	70.1	73.2	70.5
Morocco	4,108	0.654	130	71.0	55.6	61.0

Source: Data from United Nations Development Programme, Human Development Report, 2009, tab. 1.

Figure A2.1.1 Human Development Disparities Within Selected Countries



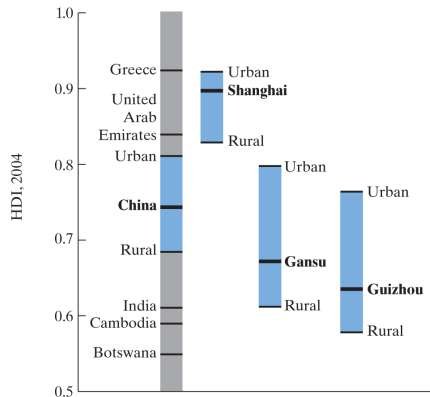
(a) Large ethnic differences in HDI in Guatemala



(b) Wide inequalities in human development between districts in Kenya

Source: From Human Development Report, 2005, figs. 10–12. Reprinted with permission from the United Nations Development Programme.

Figure A2.1.1 Human Development Disparities Within Selected Countries (Continued)



(c) Rural-urban differences intensify regional disparities in China

Source: From Human Development Report, 2005, figs. 10–12. Reprinted with permission from the United Nations Development Programme.

Appendix 2.2: How Today's Low-Income Countries Differ from Early Developers

- ▶ **Eight Relatively Widespread Aspects of Differences:**
- ▶ Physical and human resource endowments
- ▶ Per capita incomes and levels of GDP in relation to the rest of the world
- ▶ Climate
- ▶ Population size, distribution, and growth
- ▶ Historic role of international migration
- ▶ International trade benefits
- ▶ Basic scientific/technological research and development capabilities
- ▶ Efficacy of domestic institutions

Note on National Income Definitions and Calculations

- ▶ **Gross National Income (GNI)**

- ▶ Total income by resident producers + net income from abroad
- ▶ Calculated using the World Bank Atlas method (smooths exchange rates)

- ▶ **Purchasing Power Parity (PPP) GNI**

- ▶ GNI converted into international dollars
- ▶ Adjusts for price level differences across countries
- ▶ Better for comparing living standards

- ▶ **GNI per Capita**

- ▶ $\text{GNI} \div \text{midyear population}$ Used to classify country income levels

- ▶ **Gross Domestic Product (GDP)**

- ▶ Total value of goods/services produced within a country Excludes income from abroad

- ▶ **GDP per Capita**

- ▶ $\text{GDP} \div \text{midyear population}$ Rough measure of average productivity

- ▶ **Growth Measurement**

- ▶ Based on constant price GDP (adjusted for inflation)

- ▶ *For further details see "Sources and Methods," in *World Development Indicators (WDI)*, 2017

Concepts for Review

- ▶ Absolute poverty
- ▶ Brain drain
- ▶ Capital stock
- ▶ Convergence
- ▶ Crude birth rate
- ▶ Dependency burden
- ▶ Depreciation (of the capital stock)
- ▶ Diminishing marginal utility
- ▶ Divergence
- ▶ Economic institutions
- ▶ Fractionalization
- ▶ Free trade
- ▶ Gross domestic product (GDP)
- ▶ Gross national income (GNI)
- ▶ Human capital
- ▶ Human Development Index (HDI)
- ▶ Imperfect market
- ▶ Incomplete information
- ▶ Infrastructure
- ▶ Least developed countries
- ▶ Low-income countries (LICs)
- ▶ Middle-income countries
- ▶ Newly industrializing countries (NICs)
- ▶ Purchasing power parity (PPP)
- ▶ Research and development (R&D)
- ▶ Resource endowment
- ▶ Terms of trade
- ▶ Value added
- ▶ World Bank